

ABSTRACT

This project proposes a portable, low-power IoT node for real-time environmental health monitoring using the STM32 B-L4S5I-IOT01A Discovery Kit. The system is designed to deliver immediate and actionable air-quality feedback to users. It performs multi-sensor data fusion by combining on-board environmental sensors (temperature and humidity) with external sensors, including the PMS5003 for PM2.5 measurement via UART and the SGP30 for TVOC sensing via I2C.

The STM32's floating-point unit (FPU) is utilized to compute a localized Air Quality Index (AQI), which directly controls the system's outputs. Real-time AQI values are presented on an LCD display, while air-quality severity is indicated using blinking LED. To ensure portability and energy efficiency, the system employs a low-power design, periodically waking from Stop Mode using the RTC for sensor sampling. While MQTT-based cloud connectivity is planned for future expansion, the current prototype successfully demonstrates a robust real-time sensing, processing, and local visualization platform focused on enhancing user safety.

Team Name: Chipspire

Team leader: Karra Sai Siddartha

Team Members:

1. Narendhar Puttinti (UH ID: 2454090)
2. Manaswini Gundapaneni (UH ID: 2443790)
3. Shaik Irshad Ahmed (UH ID: 2443624)
4. Mukesh Musham (UH ID: 2430886)